



What “R” We Counting

Ben Arancibia



During your Open-Source adoption journey at some point a Vice President is going to come and ask you...

**“Hey – How is open-source
adoption going?”**



[.gif from here.](#)



Director of Data Science in GSK Biostatistics

**Scientific and Adoption Lead
for Open Source for GSK
R&D**

**Passionate about open-
source and teaching**

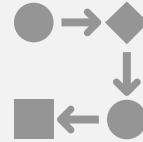
What does a Biostatistics Organization do?

Why did our VP ask us this question?

Since 2020 Putting together the Pieces



Platform



Process



Training



Tools



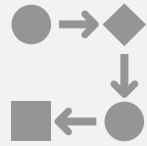
Pipeline

Interested in how we put the pieces together?

Since 2020 Putting together the Pieces



Platform



Process



Training



Tools



Pipeline



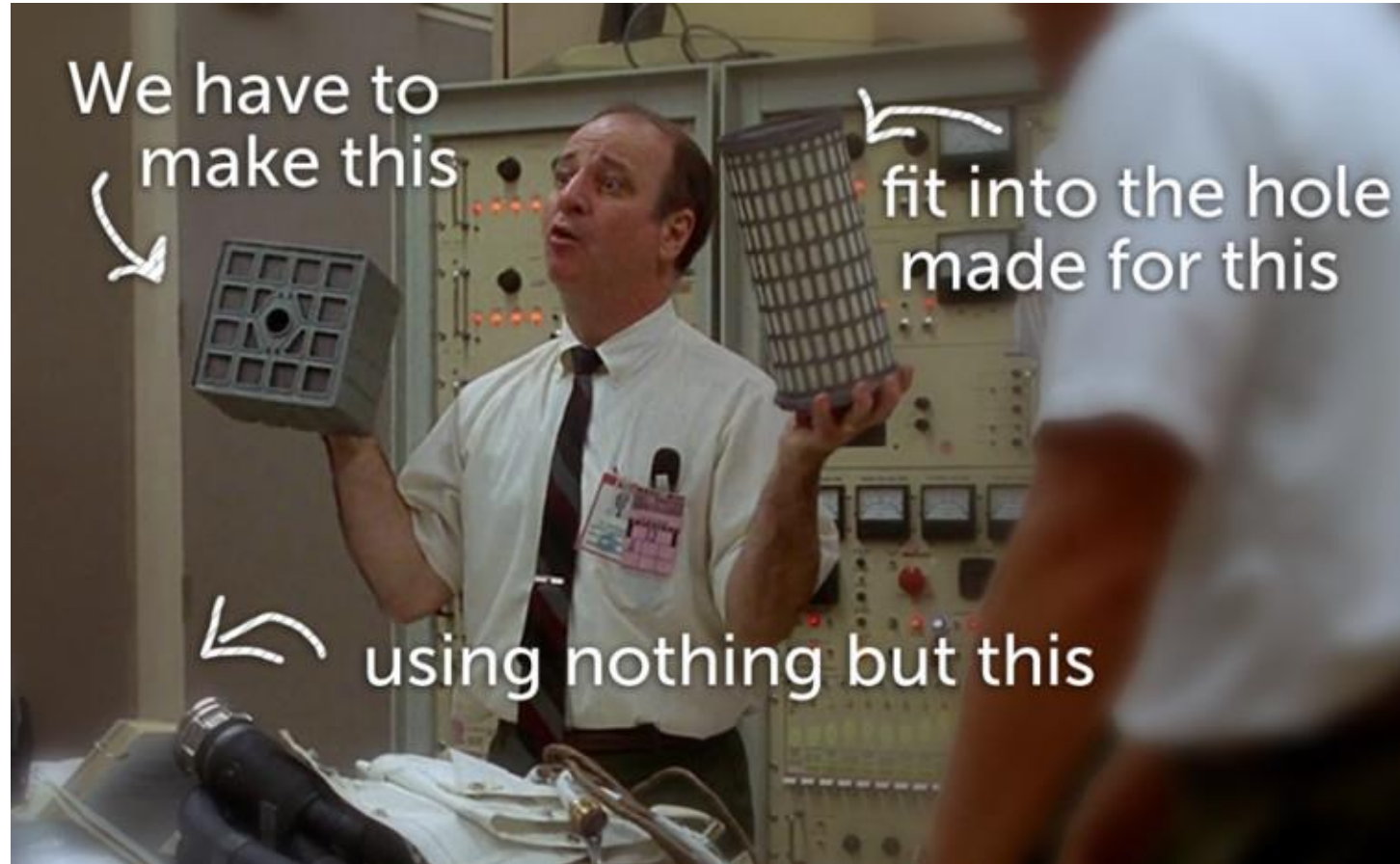
Two Commitments in August 2023

**All central tools to be built
using open-source
languages**

**50% of our code to be
written in an open-source
language by end of 2025**

Interested in how we put the pieces together?

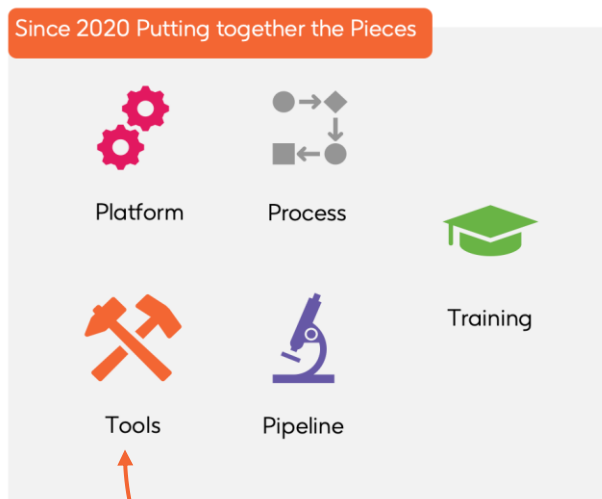
How did we answer our Vice President's question?



[Image from here.](#)



More specifically the GitHub APIs



All studies must use version control

	[skip actions] Bump version to 1.3.1.9003 ✓	75b8720 · last week	🕒 4,703 Commits
📁 .github	feat: Reintroduce multiversion docs (#2622)	2 months ago	
📁 R	[skip vbump] Closes #2788 fix_links: use vignette("vigne...	3 weeks ago	
📁 data-raw	Closes #2768 add Acidosis and Alkalosis to NCI v4 and v...	2 months ago	
📁 data	Closes #2768 add Acidosis and Alkalosis to NCI v4 and v...	2 months ago	
📁 inst	Closes #2804 ADLBHY template PARAM update (#2837)	last week	
📁 man	[skip vbump] Closes #2788 fix_links: use vignette("vigne...	3 weeks ago	
📁 pkgdown/favicon	Update favicons	3 years ago	
📁 tests	[skip vbump] Closes #2788 fix_links: use vignette("vigne...	3 weeks ago	
📁 vignettes	Closes #2780 : updates to aseq in bds finding and expos...	2 weeks ago	
📄 .Rbuildignore	Closes #2524 remove devcontainer file (#2542)	10 months ago	
📄 .gitignore	Closes #2427 general issue adopt data and data raw con...	11 months ago	
📄 .lintr	Closes #2654 derive_vars_dt error messaging (#2672)	3 months ago	
📄 .lycheeignore	Closes #2427 general issue adopt data and data raw con...	11 months ago	
📄 DESCRIPTION	[skip actions] Bump version to 1.3.1.9003	last week	
📄 LICENSE.md	License package under Apache 2.0	4 years ago	
📄 NAMESPACE	Closes #2652 derive_vars_joined_summary: Implement d...	3 months ago	
📄 NEWS.md	Closes #2804 ADLBHY template PARAM update (#2837)	last week	
📄 README.md	Closes #2752 upgraded examples for derive_var_extrem...	2 months ago	
📄 _pkgdown.yml	#2785 resize company logos	last month	
📄 admiral.Rproj	Closes #2652 derive_vars_joined_summary: Implement d...	3 months ago	
📄 staged_dependencies.yaml	admiral v0.12.0 (#2094)	2 years ago	

📖 README 🛡️ Code of conduct 🤝 Contributing 📄 Apache-2.0 license

admiral



About

ADaM in R Asset Library

pharmaverse.github.io/admiral

open-source

r

clinical-trials

cdisc

📖 Readme

📄 Apache-2.0 license

🛡️ Code of conduct

🤝 Contributing

📈 Activity

📋 Custom properties

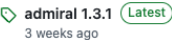
★ 252 stars

👁️ 20 watching

🍴 76 forks

Report repository

Releases 34

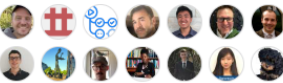
 3 weeks ago

[+ 33 releases](#)

Packages

No packages published

Contributors 95

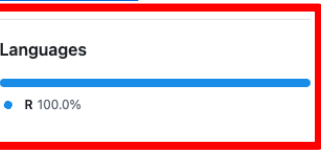


[+ 81 contributors](#)

Deployments 500+

 last week

[+ more deployments](#)



REST API endpoints for repository statistics

Use the REST API to fetch the data that GitHub uses for visualizing different types of repository activity.

About repository statistics [↗](#)

You can use the REST API to fetch the data that GitHub uses for visualizing different types of repository activity.

Best practices for caching [↗](#)

Computing repository statistics is an expensive operation, so we try to return cached data whenever possible. If the data hasn't been cached when you query a repository's statistics, you'll receive a `202` response; a background job is also fired to start compiling these statistics. You should allow the job a short time to complete, and then submit the request again. If the job has completed, that request will receive a `200` response with the statistics in the response body.

Repository statistics are cached by the SHA of the repository's default branch; pushing to the default branch resets the statistics cache.

Statistics exclude some types of commits [↗](#)

The statistics exposed by the API match the statistics shown by [different repository graphs](#).

To summarize this:

- All statistics exclude merge commits.
- Contributor statistics also exclude empty commits.

[API documentation](#).

Ask and you shall receive...

gh 1.4.1 Reference Articles Changelog

gh

Minimalistic client to access GitHub's [REST](#) and [GraphQL](#) APIs.

Installation

Install the package from CRAN as usual:

```
install.packages("gh")
```

Install the development version from GitHub:

```
pak::pak("r-lib/gh")
```

Usage

```
library(gh)
```

Use the `gh()` function to access all API endpoints. The endpoints are listed in the [documentation](#).

The first argument of `gh()` is the endpoint. You can just copy and paste the API endpoints from the documentation. Note that the leading slash must be included as well.

From <https://docs.github.com/rest/reference/repos#list-repositories-for-a-user> you can copy and paste `GET /users/{username}/repos` into your `gh()` call. E.g.

LINKS

- [View on CRAN](#)
- [Browse source code](#)
- [Report a bug](#)

LICENSE

- [Full license](#)
- [MIT](#) + file [LICENSE](#)

COMMUNITY

- [Code of conduct](#)

CITATION

- [Citing gh](#)

DEVELOPERS

- [Gábor Csárdi](#)
Maintainer, contributor
- [Jennifer Bryan](#)
Author
- [Hadley Wickham](#)
Author

posit
Copyright holder, funder

An unconventional way to use {gh}

{gh} is primarily used for managing GitHub PATs

```
install.packages("gh")
library(gh)

#Set the environment variable
Sys.setenv(GITHUB_PAT = "your_personal_access_token")

my_repos <- gh("GET /users/{username}/repos", username = "Fake Username")
vapply(my_repos, "[", "", "name")
```

BUT

It can also be used for complex queries

```
#function to get repo level information for all repos in gsk tech using graphql
get_all_repos<-function(auth = gh::gh_token()){

  # 100 repos at a time - need to paginate thru
  #(note that gh package offers limit=.Inf for REST API which paginates under the hood)
  hasNextPage <- TRUE
  endCursor <- NA
  repo_list <- vector(mode = "list")

  system.time(
    while(hasNextPage){

      query <- paste0('query {
        organization(login: "ORG NAME") {
          repositories(first: 100,
            after: ', jsonlite::toJSON(endCursor, auto_unbox = TRUE), ') {

            pageInfo {
              hasNextPage
              endCursor
            }
            edges{
              node{
                name
                diskUsage
                createdAt
                pushedAt
                updatedAt
                templateRepository{
                  name
                }
              }
            }
            # Fetch only top 10 languages
            languages(first:10){
              totalSize
              edges {
                size
                node{
                  name
                }
              }
            }
          }
        }
      }')

      repo_list[[length(repo_list)+1]] <- gh(repo_list, auth = auth)
    }
  )
}
```

**Started exploring and pulling data
through the API**

Count number of outputs in a study?

Do you count

Do you care
languages?

Do you timebox it?

is or
both?

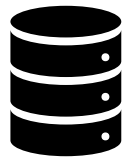
matter

So what do you measure....



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Relative R Size



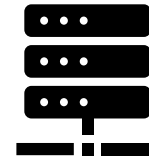
R code size out of total
(R and SAS)

Any R



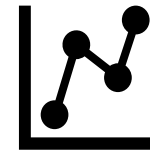
Total Repos with any
usage of R

Substantial R



Repos where R code
size (out of R and SAS)
is $\geq$ Target Percent

Growth of R



Rate of change of R
code in a repo and
organization



Select a Therapeutic Area:

Overall R Adoption



Select a Disease Area:

All Disease Areas



Select a Project ID:

All Projects



Select a Study:

Overall R Adoption



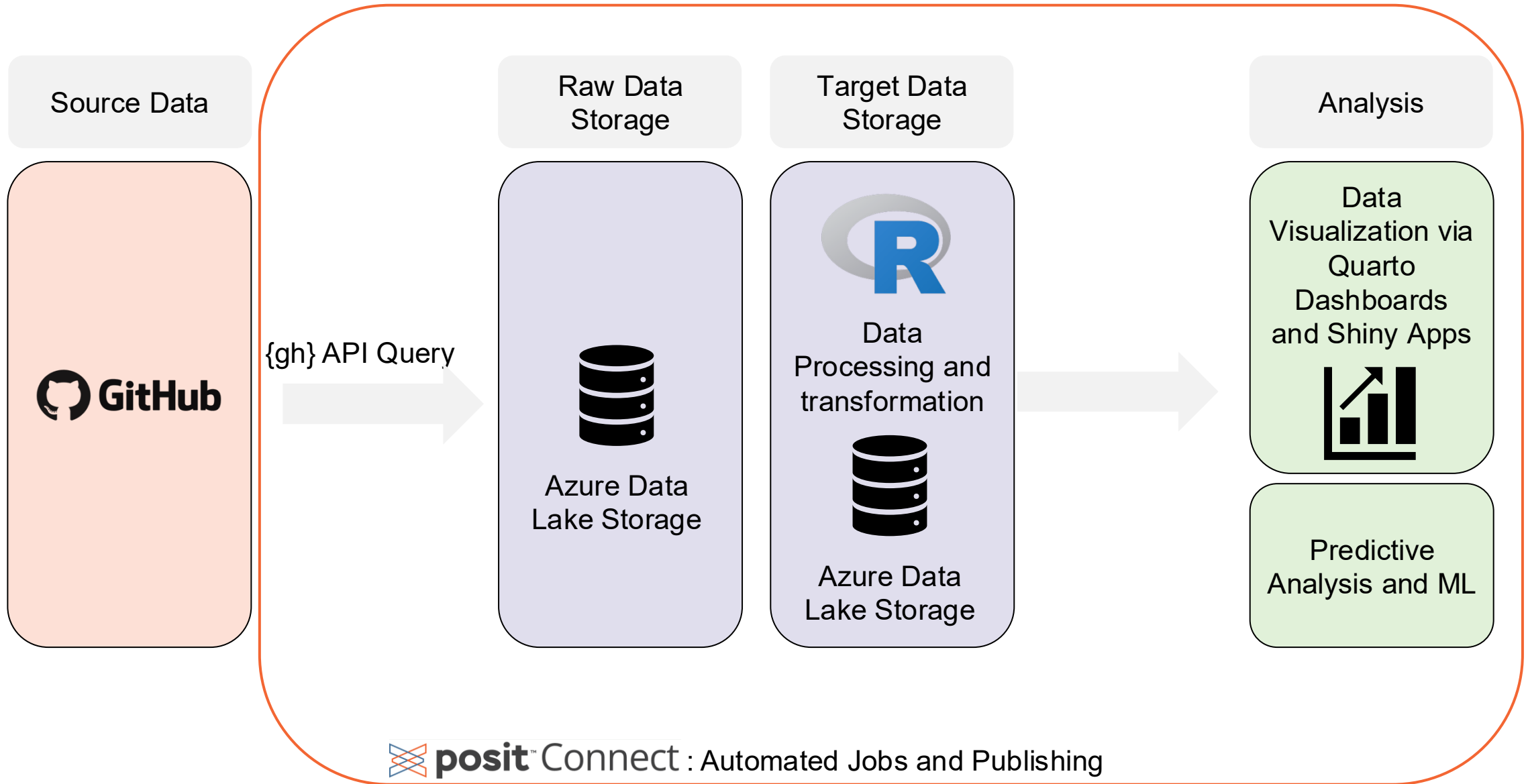
QC

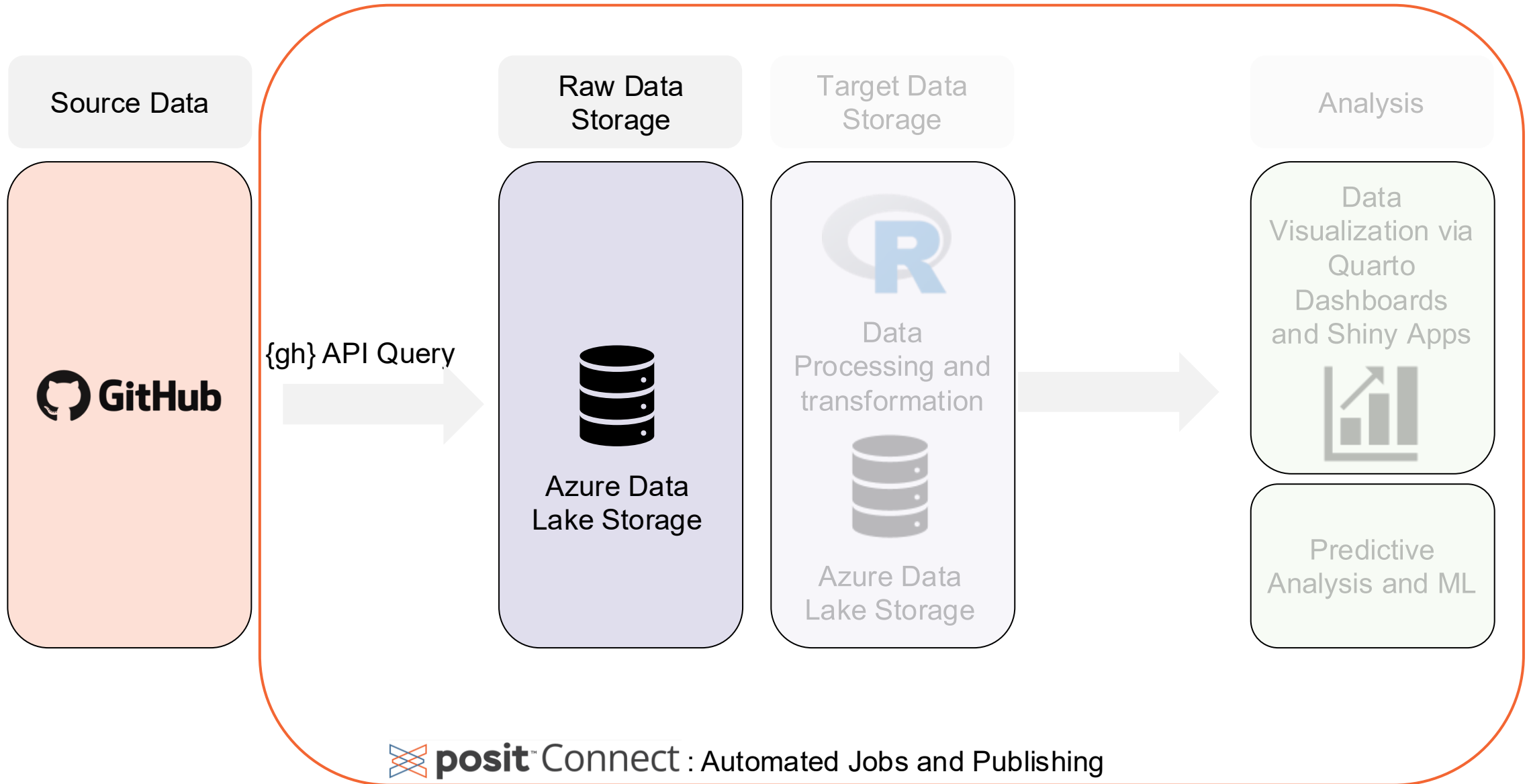
Production

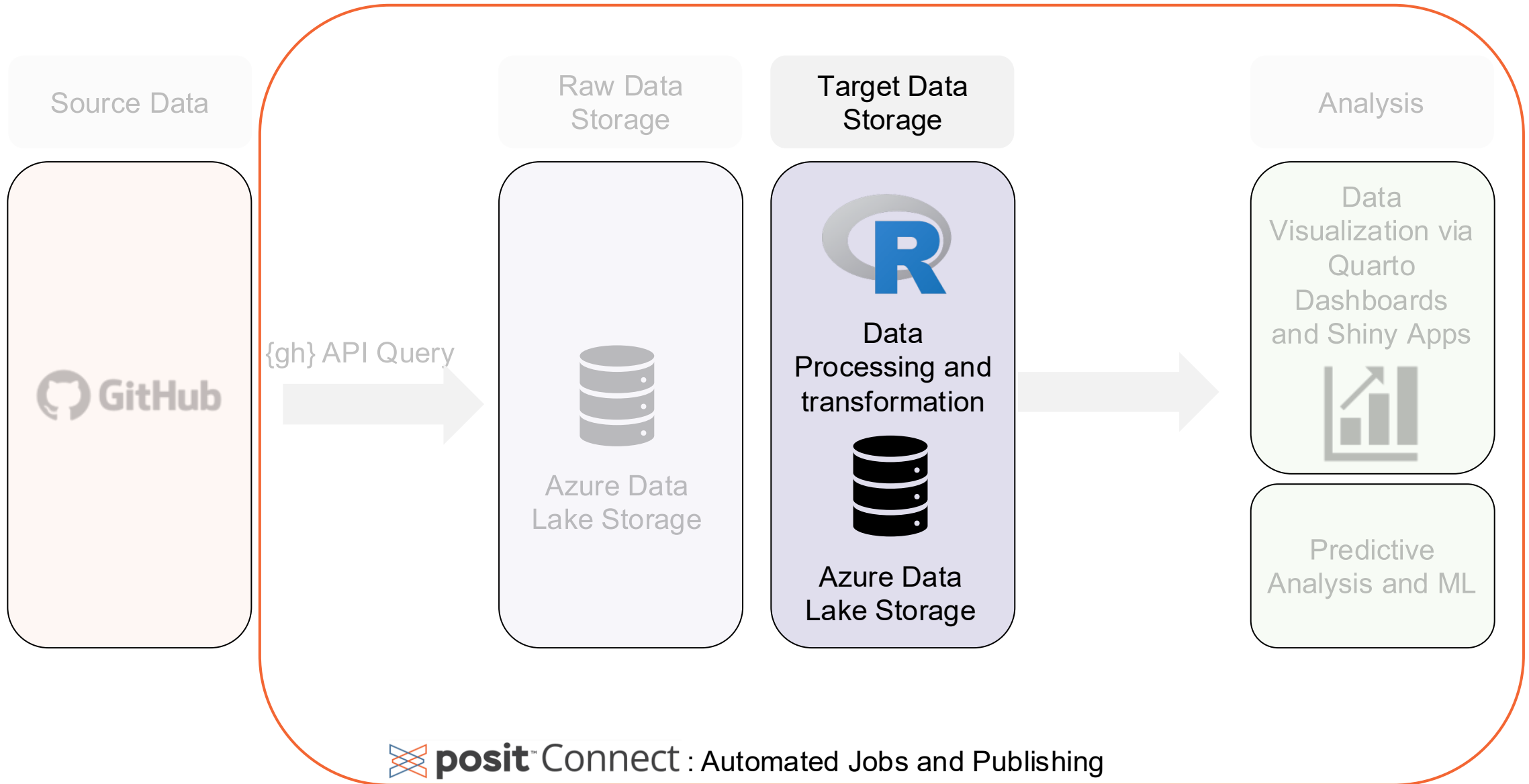
Fancy Metric Here!

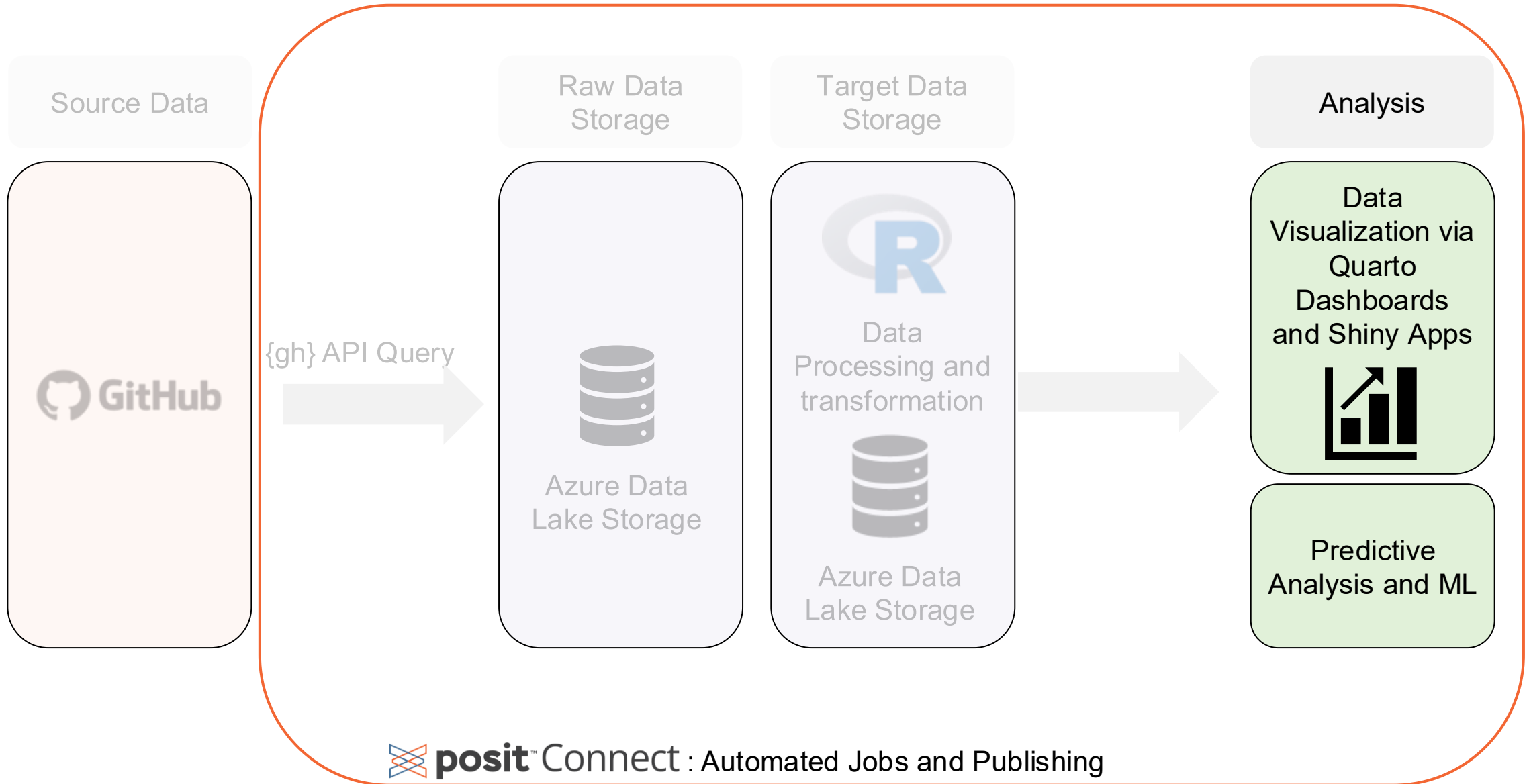
Fancy Metric Here!

Fancy Metric Here!











[.gif from here.](#)



.gif from here.

Useful Data

- Commits
- Pull Requests
- Merge Conflicts
- Number of Branches
- Repo Life (days)
- Repo Pull Request Reviewers
- Repo Names
- Studies with multiple repos
- Irregular commits
- Unmerged pull requests
- Package Names

GitHub Health

GitHub is hard to learn...how do we support users in this transition.

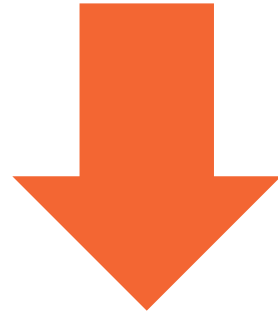
Can we proactively find studies that are struggling and support them?

Tool Catalogue

We spend a lot of time building tools...do our users actually use them?

What tools are being used? What's our return on investment (ROI) for internal tools and open-source tools?

“Hey how is open-source adoption going?”



“I know exactly how adoption is going”

It takes a team:

Becca Krouse

Alanah Jonas

Niladri Dasgupta

Hamza Ferhusoglu

Hashir Anhar



[.gif from here.](#)

Thank You!
Any Questions?

GSK

SCAN ME



Feel free to connect!

<https://www.linkedin.com/in/bcarancibia/>